

Independent Section

Contains tests that are independent of the class of modeled organism, a model's complexity or types of identifiers that are used to describe its components. Parameterization or initialization of the network is not

Consistency

Stoichiometric Consistency	100.0%	x3
Mass Balance	74.5%	
Charge Balance	92.1%	
Metabolite Connectivity	100.0%	
Unbounded Flux In Default Medium	99.1%	
Sub Total	95%	x3

Annotation - Metabolites

Presence of Metabolite Annotation	100.0%	
Metabolite Annotations Per Database	Info	
pubchem.compound	68.5%	
kegg.compound	81.6%	
seed.compound	0.0%	
inchikey	72.7%	
inchi	72.7%	
chebi	64.9%	
hmdb	0.0%	

Specific Section

Covers general statistics and specific aspects of a metabolic network that are not universally applicable. See readme for more details.

SBML

SBML Level and Version	SBML Level 3 Version 1	
FBC enabled	true	

Basic Information

Model Identifier	A_28316_5_10	
Total Metabolites	359	
Total Reactions	370	
Total Genes	300	
Total Compartments	2	
Metabolic Coverage	1.23	
Unconserved Metabolites	0	
Minimal Inconsistent Net Stoichiometries	0	

Metabolite Information

Unique Metabolites	315	
Duplicate Metabolites in Identical Compartments	0	

reactome	0.0%	▼
metanetx.chemical	85.2%	▼
bigg.metabolite	82.7%	▼
biocyc	78.6%	▼

Metabolite Annotation Conformity Per Database

	Info	▼
pubchem.compound	100.0%	▼
kegg.compound	97.6%	▼
seed.compound	0.0%	▼
inchikey	100.0%	▼
inchi	100.0%	▼
chebi	100.0%	▼
hmdb	0.0%	▼
reactome	0.0%	▼
metanetx.chemical	99.7%	▼
bigg.metabolite	100.0%	▼
biocyc	98.6%	▼
Uniform Metabolite Identifier Namespace	100.0%	▼

Sub Total	82%	▼
-----------	-----	---

Annotation - Reactions

Presence of Reaction Annotation	100.0%	▼
Reaction Annotations Per Database	Info	▼
rhea	56.5%	▼

Metabolites without Charge	0	▼
Metabolites without Formula	53	▼
Medium Components	47	▼

Reaction Information

Purely Metabolic Reactions	274	▼
Purely Metabolic Reactions with Constraints	19	▼
Transport Reactions	44	▼
Transport Reactions with Constraints	1	▼
Reactions With Partially Identical Annotations	0.03	▼
Duplicate Reactions	0.00	▼
Reactions With Identical Genes	0.41	▼

Gene-Protein-Reaction (GPR) Associations

Reactions without GPR	1	▼
Fraction of Transport Reactions without GPR	0.00	▼
Enzyme Complexes	Errored	▼

Biomass

Biomass Reactions Identified	1	▼
Biomass Consistency	Errored	▼
Biomass Production In Default Medium	3.99	▼
Unrealistic Growth Rate In Default Medium	true	▼

kegg.reaction	63.2%	▼
seed.reaction	0.0%	▼
metanetx.reaction	77.0%	▼
bigg.reaction	67.6%	▼
reactome	0.0%	▼
ec-code	0.0%	▼
brenda	0.0%	▼
biocyc	60.0%	▼

Reaction Annotation Conformity Per Database

Info

rhea	100.0%	▼
kegg.reaction	100.0%	▼
seed.reaction	0.0%	▼
metanetx.reaction	100.0%	▼
bigg.reaction	99.5%	▼
reactome	0.0%	▼
ec-code	0.0%	▼
brenda	0.0%	▼
biocyc	100.0%	▼
Uniform Reaction Identifier Namespace	100.0%	▼

Sub Total	73%	▼
-----------	-----	---

Annotation - Genes

Presence of Gene Annotation	100.0%	▼
-----------------------------	--------	---

Biomass Production In Complete Medium	3.99	▼
Blocked Biomass Precursors In Default Medium	2	▼
Blocked Biomass Precursors In Complete Medium	2	▼
Ratio of Direct Metabolites in Biomass Reaction	0.32	▼
Number of Missing Essential Biomass Precursors	1	▼

Energy Metabolism

Non-Growth Associated Maintenance Reaction	0	▼
Growth-associated Maintenance in Biomass Reaction	true	▼
Number of Reversible Oxygen-Containing Reactions	1	▼

Erroneous Energy-generating Cycles

Info

MNXM3	331	▼
MNXM63	8	▼
MNXM51	4	▼
MNXM121	8	▼
MNXM423	Skipped	▼
MNXM6	0	▼
MNXM10	0	▼
MNXM38	0	▼
MNXM208	Skipped	▼
MNXM191	0	▼
MNXM223	Skipped	▼
MNXM7517	Skipped	▼

Gene Annotations Per Database

Info

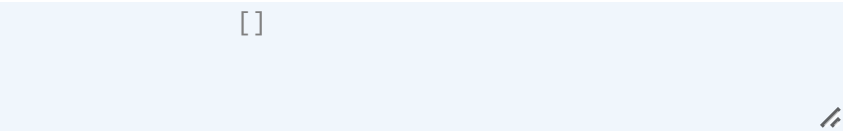
refseq	91.0%	
uniprot	88.7%	
ecogene	0.0%	
kegg.genes	91.0%	
ncbigi	0.0%	
ncbigene	10.7%	
ncbiprotein	91.0%	
ccds	0.0%	
hprd	0.0%	
asap	0.0%	

Gene Annotation Conformity Per Database

Info

refseq	100.0%	
uniprot	98.7%	
ecogene	0.0%	

There are no gene annotations for the ecogene database.



kegg.genes	100.0%	
ncbigi	0.0%	
ncbigene	100.0%	
ncbiprotein	100.0%	
ccds	0.0%	

MNXM12233	Skipped	
MNXM558	Skipped	
MNXM21	10	
MNXM89557	0	

Network Topology

Universally Blocked Reactions	13	
Orphan Metabolites	3	
Dead-end Metabolites	1	
Stoichiometrically Balanced Cycles	5	
Metabolite Production In Complete Medium	55	
Metabolite Consumption In Complete Medium	62	

Matrix Conditioning

Ratio Min/Max Non-Zero Coefficients	0.00	
Independent Conservation Relations	10	
Rank	349	
Degrees Of Freedom	21	

Experimental Data Comparison

Growth Prediction	Skipped	
Gene Essentiality Prediction	Skipped	

Misc. Tests

hprd	0.0%	▼
asap	0.0%	▼

Sub Total	62%	▼
-----------	-----	---

Annotation - SBO Terms

Metabolite General SBO Presence	100.0%	▼
Metabolite SBO:0000247 Presence	100.0%	▼
Reaction General SBO Presence	100.0%	▼
Metabolic Reaction SBO:0000176 Presence	100.0%	▼
Transport Reaction SBO:0000185 Presence	100.0%	▼
Exchange Reaction SBO:0000627 Presence	100.0%	▼
Demand Reaction SBO:0000628 Presence	Skipped	▼
Sink Reactions SBO:0000632 Presence	Skipped	▼
Gene General SBO Presence	100.0%	▼
Gene SBO:0000243 Presence	100.0%	▼
Biomass Reactions SBO:0000629 Presence	100.0%	▼

Sub Total	82%	x2 ▼
-----------	-----	---------

Total Score	85%	▼
-------------	-----	---

Total Score

Environment

Python Version	3.12.7
Platform	Linux
Memote Version	

85%

Score per Category

